SAE S1.03

Installation of a development workstation.

1. Project overview:

This project involves setting up a development workstation by installing Linux using two methods: a virtual machine running on Windows and a dual-boot installation. We must select and configure the tools, enable file sharing, and then verify the workstation's operation.

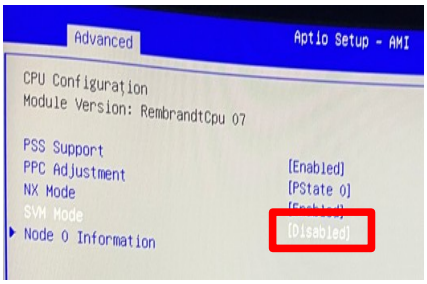
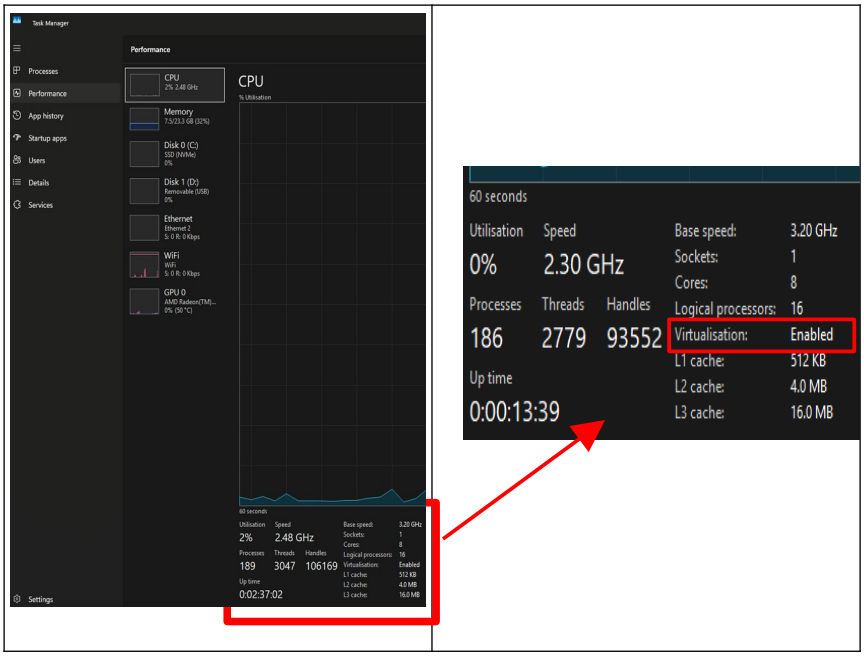
2. Hardware and software:

- Host system: Windows 11 Pro (64-bit)
- Processor: AMD Ryzen 7 7735HS, 8 cores and 16 logical processors, 3.20 GHz base frequency
- Random Access Memory (RAM): 24 GB DDR5
- Storage: 512 GB NVMe SSD (477 GB usable)
- Available disk space: 14 GB
- Virtualisation software: Oracle VirtualBox, version 7.2.20
- Guest operating system: Ubuntu Desktop 26.04.1 LTS (AMD64)
- Storage media used: a 32 GB USB flash drive formatted in exFAT, containing the Ubuntu ISO image and the VirtualBox installer.
- Administrator rights: available
- Hardware virtualisation: enabled

3. Virtual machine installation (individual work):

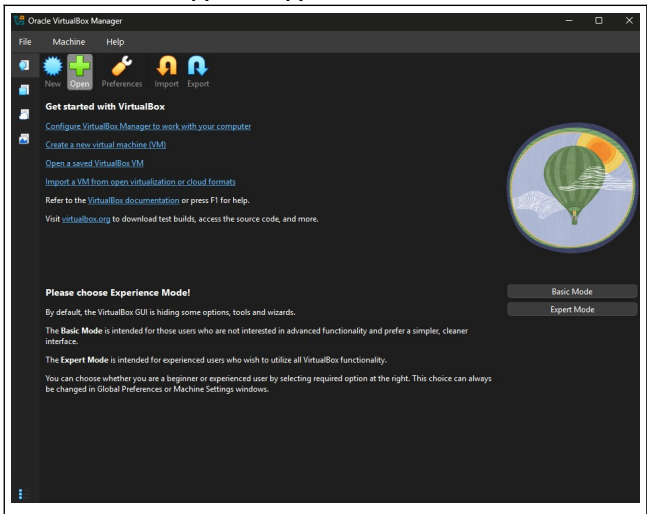
Step 1 — Check the PC configuration:

We note the PC's specifications and check for the availability of hardware virtualization in Task Manager.

If virtualisation is disabled, we enable it in the UEFI/BIOS and then restart Windows. 	*We note the PC's specifications and check for the availability of hardware virtualisation in **Task Manager**, under **Performance → CPU**.* 

Step 2 — Install VirtualBox (Oracle) and download VirtualBox and the Linux (Ubuntu) ISO image.

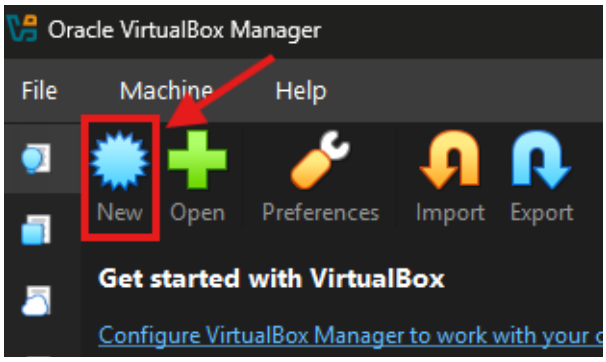
We launch the VirtualBox installation on Windows and complete the installation wizard. This software is a Type 2 hypervisor.



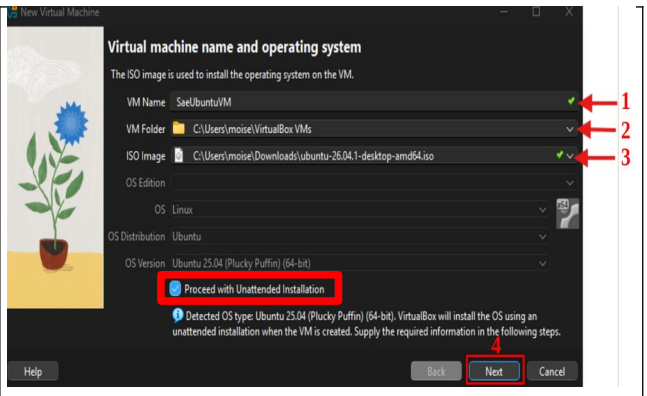


Step 3 — Create the virtual machine

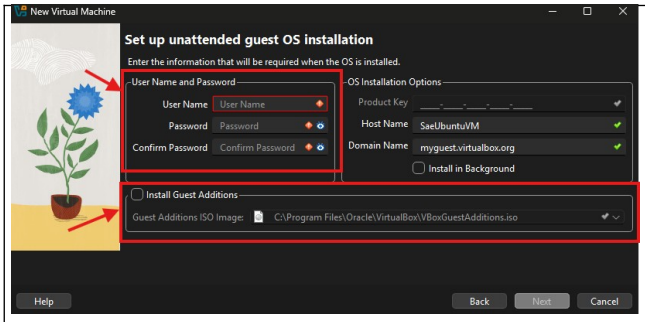
In VirtualBox, we create a new machine of the Linux/Ubuntu type and select the Ubuntu ISO.



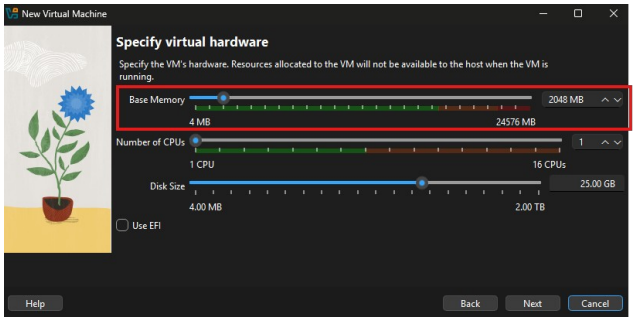
*We open Oracle VirtualBox and click "**New**" to create a new virtual machine in which we will install Ubuntu.*



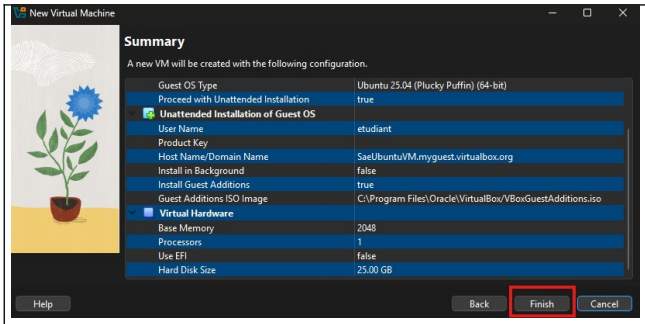
1. *Name your virtual machine.*
2. *Choose the location you wish to assign to it.*
3. *Select the ISO image we just downloaded.*
4. *Then proceed to the next step, but first check that the "**Proceed with unattended Installation**" option is selected.*



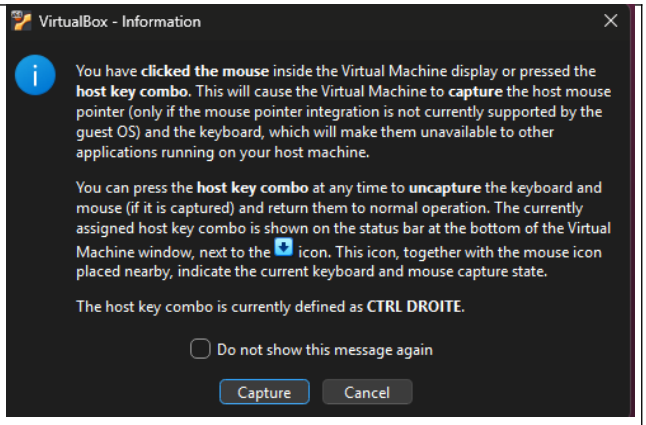
*We set a username and password to access Ubuntu. We keep the hostname and domain suggested by VirtualBox, then enable "**Install Guest Additions**" to facilitate integration between Windows and Ubuntu, particularly file sharing.*



We configure the amount of RAM, the number of processors, and the virtual hard disk size. These resources are allocated to Ubuntu based on the computer's capabilities, while reserving sufficient resources for Windows.



*VirtualBox displays a summary of the selected settings: operating system, user account, Guest Additions, and hardware resources. We verify this information before clicking "**Finish**" to create the virtual machine and start the installation.*



When the virtual machine starts up, VirtualBox displays a message regarding mouse and keyboard capture. We click "Capture" to interact with Ubuntu. The right Ctrl key allows you to release the mouse and keyboard and return to Windows.